SmartUzhavan

கிராம கணக்கு மேலாண்மை அமைப்பு

PROJECT DOCUMENTATION — VERSION 3.0

Full System Audit | Architecture Review | Bug Analysis | Action Plan

Project	SmartUzhavan — Rural Ledger & Machinery System
Version	3.0 — Post v2 Correction Patch Audit
Architecture	React (Vite) Frontend + Node.js PDF Backend
Issues Found	**14 issues (3 Crash-level, 6 Logic errors, 5 Minor)**
Date	April 2026

1. Project Overview

SmartUzhavan is a Tamil-first, offline-first, mobile-first agricultural business management system built for a rural entrepreneur in Tamil Nadu. Version 3.0 represents the system after the v2 correction patch — a significant upgrade from v1 that resolved authentication, added new modules, and migrated PDF generation to a Node.js backend.

1.1 What Changed from v1 to v2 (Before This Audit)

Module / Feature	v1 Status	v2 Status
Authentication (Login)	❌ Completely missing	✔️ Implemented — role-based with PIN
Farmer View Portal	❌ Completely missing	✔️ Implemented with payment history
Finance Ledger (Lending)	❌ Completely missing	✔️ Implemented with repayments[]
Finance Ledger (Home Expenses)	❌ Missing	⚠️ Stub — redirects to Expenses page
Reports Module	❌ Completely missing	✔️ 5 report types implemented
Settings (Backup/Restore)	❌ Missing	✔️ Implemented (no schemaVersion yet)
Driver Dashboard	❌ Missing	✔️ New DriverDashboard.jsx added
Worker Net Payable Formula	❌ Wrong formula	✔️ Fixed — baseSalary+bonus+extra-advance
Driver Salary Logic	❌ All-time overcounting bug	✔️ Fixed — linkedSalaryId pattern
Harvester otherExpenses UI	❌ Missing	✔️ Dynamic add/remove expenses
Harvester date field	❌ Missing	✔️ date field added to jobs
Own Farm Income schema	❌ Wrong schema	✔️ Corrected to numberOfBags × pricePerBag
Dashboard month filter	❌ All-time totals	✔️ Month/year selector added
Dashboard job expenses	❌ Not counted	✔️ dieselFromLogs + otherExpenses counted
PDF Generation	❌ Frontend (broken Tamil)	✔️ Node.js backend with PDFKit
Tamil font in PDF	⚠️ 81-byte placeholder	✔️ Real NotoSansTamil.ttf (322KB)

2. System Architecture

2.1 Frontend (React + Vite)

The frontend is a Single-Page Application (SPA) built on React 19 with Vite 8. All business data is stored in browser LocalStorage. The frontend handles all user interactions, calculations, and data management. PDF generation is delegated to the backend via HTTP API calls.

Layer	Technology	Purpose
UI Framework	React 19 (Vite 8)	Component rendering, state management, routing
Routing	react-router-dom v7	Role-based page routing with in-memory guards
Data Storage	Browser LocalStorage	All 11 business data collections
Authentication	localStorage('su_session')	Role session (admin/driver/farmer)
PDF Requests	fetch() to localhost:5000	Sends JSON, receives PDF blob
PWA	vite-plugin-pwa	Offline app install via Chrome Add-to-Home-Screen

2.2 Backend (Node.js PDF Server)

A separate Express.js server runs on port 5000. It receives JSON data from the frontend, generates PDFs using PDFKit with the NotoSansTamil font, and returns binary buffers. This architecture was chosen after multiple frontend PDF library failures with Tamil Unicode rendering.

Component	Details	Status
Framework	Express.js 5.x	Running
PDF Engine	PDFKit 0.18	Working
Tamil Font	NotoSansTamil.ttf (322KB)	✔️ Real font — verified valid
Font Safety Guard	isFontValid() checks > 10KB	✔️ Prevents crashes from stubs
Endpoints	POST /api/pdf/harvester-bill	✔️ Working
Endpoints	POST /api/pdf/statement	✔️ Working
Endpoints	POST /api/pdf/rental-receipt	❌ CRASH — import missing in index.js
CORS	cors() middleware	Allows localhost:3000/5173 frontend
Startup	No start script in package.json	⚠️ Must run: node index.js manually

2.3 Data Flow Architecture

```
USER ACTION (Frontend)
        │
        ▼
React Component  →  services/storage.js  →  LocalStorage
                        (getData / addRecord / updateRecord)
                              │
                              ▼
                     services/calculations.js
                     (computed values — never stored)
                              │
                              ▼
```

```
                        React State  →  UI Re-render

PDF REQUEST:
React Component  →  services/pdfService.js
    → fetch('http://localhost:5000/api/pdf/...')
    → JSON body (job/farmer data)
    ← PDF Binary Blob
    → Browser downloads file

OFFLINE CAPABILITY:
  PWA caches app shell after first load.
  LocalStorage data is always available offline.
  ⚠️ PDF generation requires backend to be running (NOT offline-capable).
```

2.4 LocalStorage Collections

Key	Type	Contents
rl_farmers	Array	Farmer records including F000 (Own Farm)
rl_drivers	Array	Driver records with salaryRatePerHour and PIN
rl_workers	Array	Worker records (name, id — no phone/village saved)
rl_driver_logs	Array	Machine usage logs with sessions, diesel, linkedSalaryId
rl_harvester_jobs	Array	Harvester billing jobs with payments[], Bill IDs
rl_rentals	Array	Machinery rental entries with payments[]
rl_driver_salary	Array	Driver salary records linked to logs, with payments[]
rl_work_entries	Array	Worker work entries with netPayable formula fields
rl_expenses	Array	General/own-farm expenses (no source field — issue)
rl_own_farm_income	Array	Own farm harvest income (bags × price computed)
rl_lending	Array	Informal lending records with repayments[]
rl_pricing_config	Object	All machine rates and diesel price per litre
su_session	Object	Active login session {role, id, name}

2.5 Package.json — Dead Dependencies Warning

The frontend package.json still includes jsPDF, html2pdf.js, and pdfmake as dependencies.
These libraries were used during the failed frontend PDF attempts and are now replaced by the backend.
They are loaded into the browser bundle unnecessarily, increasing app load time.

Recommended: Remove all three from package.json and run npm install.

3. Module Breakdown — Detailed Audit

3.1 Authentication Module

File: src/pages/Login.jsx | Session: localStorage('su_session')

✔ WORKS	**Role-based login with driver PIN verification and farmer no-password access**

Three login roles are implemented. Admin uses a hardcoded PIN '1234'. Drivers select their name from a dropdown and enter their 4-digit PIN (which is stored in the driver record). Farmers select their name — no PIN needed. On successful login, role session is stored in localStorage.

Aspect	Status	Detail
Admin login	✔ Works	PIN: '1234' hardcoded — cannot be changed by user
Driver login	✔ Works	Validates against driver.pin; fallback '0000' also accepted
Farmer login	✔ Works	Name selection only, no password
Session persistence	✔ Works	Survives page reload via localStorage
Route protection	✔ Works	App.jsx renders only routes matching user.role
Logout	✔ Works	Clears su_session and resets user state
Admin PIN changeable	❌ Missing	PIN is hardcoded as '1234' in source code
Driver fallback '0000'	⚠ Issue	Any driver can log in with PIN '0000' regardless of their actual PIN

3.2 Driver Module

Files: DriverEntry.jsx, DriverDashboard.jsx, SalaryDriver.jsx

DriverEntry.jsx

⚠ ISSUE	**Core entry works but has 3 problems**

- ✔ Sessions with overnight guard — time calculation is correct
- ✔ Diesel modes (none/rupees/litres) all working
- ✔ Driver is pre-set from userId when logged in as driver role
- ✔ Today's logs shown below form after save
- ✔ generateLogBillId() assigns a Bill ID to each log

ISSUE 1 — Default machineType is 'tyre' (invalid enum value):

Initial state: machineType: 'tyre'
Valid options in machineTypes.js: 'harvester_tyre', 'harvester_track', 'tractor', 'plough_tractor', 'trailer'
If driver saves without changing the machine type, the log is saved with machineType: 'tyre'
which is not a valid enum value and will cause filtering issues in the Harvester module.
Fix: Change initial value to 'harvester_tyre'.

ISSUE 2 — Default sessions pre-filled as 09:00–10:00:
New sessions default to 09:00 start and 10:00 end (1 hour).
If a driver saves without changing the times, incorrect hours are recorded silently.
Fix: Default sessions to { start: '', end: '', durationHours: 0 }.

ISSUE 3 — generateLogBillId hardcodes season as '01':
const ss = '01'; // hardcoded
This creates Bill IDs like 'Bill-2026-01-001' regardless of actual season.
This is cosmetic but misleading. The SRS does not define Bill IDs for driver logs —
only for Harvester Jobs (HB-SSS-YYYY-NNN). Consider removing log Bill IDs
or making them clearly sequential without season context.

DriverDashboard.jsx

✔ WORKS	**Correctly shows hours worked, salary earned, received, and pending balance**

The Driver Dashboard correctly filters data by userId. It shows total hours worked (all-time), days worked (unique dates), total earned (from netPay in salary records), total received (from payments[] in salary records), and the difference as pending salary. The breakdown of advance, bonus, and extra amounts is clearly displayed.

Minor note: Driver Dashboard shows ALL-TIME stats without a month filter. This is acceptable for a driver view (they want to see total outstanding balance), but consider adding a month selector in a future version for detailed tracking.

SalaryDriver.jsx

✔ WORKS	**Correctly uses linkedSalaryId to find unpaid logs; partial payments via payments[]**

The salary module correctly finds all driver logs that do NOT have a linkedSalaryId field. These are the unpaid hours. Base salary is computed as totalHours × salaryRatePerHour. After saving, each processed log is updated with the salaryId. Partial payments are recorded through the payments[] array with PaymentHistory component. Drivers can view their own salary history in read-only mode when logged in.

Remaining issue: Success confirmation uses alert() in English: 'Salary record created successfully!'
This should be a Tamil inline success message following the rest of the app's pattern.

Fix: Replace alert() with setSuccess state and show Tamil toast.

3.3 Farmer Module

File: FarmerView.jsx

✔ WORKS	**Farmer dashboard, bill details, and payment history all working**

The Farmer View portal correctly filters all Harvester Jobs and Rental Entries by the logged-in farmer's userId. The dashboard shows current season paid amount and overall outstanding balance. Bills and rentals are sorted by date. Each bill expands to show individual payment records. PDF buttons trigger backend API calls for harvester bills, rental receipts, and full account statements.

Feature	Status	Note
Farmer data isolation	✔ Correct	Only own farmerId records shown
Total outstanding balance	✔ Correct	Sum across all jobs and rentals
Payment history per bill	✔ Correct	Expandable with date/mode display
Harvester Bill PDF	✔ Working	Calls /api/pdf/harvester-bill
Account Statement PDF	✔ Working	Calls /api/pdf/statement
Rental Receipt PDF	❌ Crashes backend	generateRentalReceipt not imported in server/index.js
Date sorting	⚠ Partial	Falls back to item.createdAt which doesn't exist — uses 'Invalid Date'
F000 Own Farm bills	✔ Correctly excluded	Own Farm jobs are never shown to farmers

3.4 PDF Generation System

Files: server/index.js, server/pdfGenerator.js, src/services/pdfService.js

⚠ ISSUE	**2 out of 3 endpoints work; rental receipt crashes; localhost URL will break in production**

What Works

- ✔ NotoSansTamil.ttf is 322KB — a real, valid font. Tamil text will render correctly.
- ✔ isFontValid() safety guard prevents server crash from stub fonts.
- ✔ Harvester Bill PDF — complete layout with farmer details, bill info, payments.
- ✔ Account Statement PDF — summary boxes, transaction table, totals.

- ✔️ pdfService.js correctly handles blob download and cleanup.

Critical Bug 1 — Rental Receipt Crashes the Server

```
server/index.js imports:
  const { generateHarvesterBill, generateStatement } = require('./pdfGenerator');

generateRentalReceipt is defined in pdfGenerator.js BUT NOT destructured in this import.

The rental-receipt endpoint then calls generateRentalReceipt(rental, farmer);
which throws: ReferenceError: generateRentalReceipt is not defined

This crashes the entire server process.

Fix (1 line in server/index.js):
  const { generateHarvesterBill, generateStatement, generateRentalReceipt }
    = require('./pdfGenerator');
```

Critical Bug 2 — Hardcoded localhost URL Breaks in Production

```
src/services/pdfService.js:
  const API_BASE_URL = 'http://localhost:5000/api/pdf';

This works only when both frontend and backend run on the same machine.
When the frontend is deployed to Vercel (or any hosting):
  — The backend is NOT on localhost:5000 on the user's phone
  — Every PDF button will fail with a network error

For the current local-only use case this is acceptable.
Before any deployment, API_BASE_URL must be updated to the backend server URL,
or an environment variable must be used:
  const API_BASE_URL = import.meta.env.VITE_PDF_API_URL || 'http://localhost:5000/api/pdf';
```

Rental Receipt PDF Content Bug

```
In pdfGenerator.js, the generateRentalReceipt function uses:
  rental.equipmentName   — this field does not exist in rental records
  rental.duration        — this field does not exist in rental records
  rental.durationUnit    — this field does not exist in rental records

Actual rental record fields are: machineType, unit, quantity, ratePerUnit, totalAmount

Fix: Replace references in the PDF template:
  rental.equipmentName  →  rental.machineType
  rental.duration       →  rental.quantity
  rental.durationUnit   →  rental.unit
```

3.5 Admin Module

Files: Dashboard.jsx, Harvester.jsx, Workers.jsx, Finance.jsx, Expenses.jsx, Reports.jsx, Settings.jsx

Dashboard.jsx

⚠️ ISSUE **Month filter works for most modules but Harvester jobs are NOT filtered by month**

```
// CURRENT CODE — Bug:
const filteredJobs = jobs.filter(j => j.status !== 'cancelled');
// ⚠️ filterByDate is NOT applied to jobs. ALL active jobs are summed regardless of
month.

// CORRECT CODE:
const filteredJobs = jobs.filter(j =>
  j.status !== 'cancelled' && filterByDate(j.date)
);
```

This means the dashboard income figure for harvester is always ALL-TIME total, regardless of which month is selected. Selecting April 2026 will still show all harvester income from every month. This is a significant accuracy issue.

Metric	Filter Applied?	Correct?
Harvester Revenue	❌ NOT filtered by month	Shows all-time total — WRONG
Rental Revenue	✔️ Filtered by rental.date	Correct
Own Farm Revenue	✔️ Filtered by harvestDate	Correct
Driver Salaries	✔️ Filtered by date	Correct
Worker Wages	✔️ Filtered by date	Correct
General Expenses	✔️ Filtered by date	Correct
Job-level Expenses	❌ NOT filtered by month	Follows unfiltered jobs — WRONG
Pending Bills Count	Uses all jobs	Correct (all-time is intentional)
Farmer Dues Balance	Uses all jobs+rentals	Correct (all-time is intentional)
Unpaid Driver Salary	Hardcoded to 0	Not implemented — shows placeholder

Harvester.jsx

✔ WORKS	**Well-implemented with otherExpenses, date field, log-linking guard, and payments[]**

The Harvester module is the most well-implemented complex module in the system. It correctly handles: date field on jobs (needed for dashboard filtering), otherExpenses dynamic rows with label + amount, log-linking with already-linked guard (linkedIds Set), rate auto-fetched from pricing config with wet-field toggle, and payment history via PaymentHistory component.

Minor issue: alert() used for validation ('Please link at least one driver log').
Should be replaced with an inline error message in Tamil.
Note: Log filter uses .includes() check: (log.machineType||'').includes(formData.machineType)
This works correctly for 'tyre'→'harvester_tyre' and 'track'→'harvester_track' matching.
However it is fragile — any new machine type containing 'tyre' would incorrectly match.

Workers.jsx

⚠ ISSUE	**Formula is correct but schema uses workerName (string) instead of workerId (reference)**

The net payable formula is now correctly implemented: (baseSalary + bonus + extraAmount) - advance, with a Tamil warning for negative values. However the form saves workerName as a freetext string rather than workerId as a reference. This means if a worker is renamed, historical entries will show the old name permanently.

Issue	Impact	Fix
workerName (string) not workerId (reference)	Cannot look up worker details, rename cascades fail	Change to workerId, look up name on render
Initial worker seeding uses addRecord twice	Works but creates workers without phone/village	Add phone/village to initial seed
No 'Other' work type free-text option	Limited to dropdown choices	Add 'other' option with text input

Finance.jsx

⚠ ISSUE	**Lending tab works correctly; Home Expenses tab is a non-functional stub**

The Lending sub-module is correctly implemented with person name, amount, date, repayments[] via PaymentHistory, and status auto-derived from payment amounts. Outstanding issues exist with the Home Expenses sub-module.

Home Expenses tab in Finance.jsx shows only:
'Note: Home expenses are managed in the Expenses module with source=home.'
with a redirect button to /expenses

Problems:
1. Expenses.jsx does NOT have a 'source' field — no record is ever saved with source='home'
2. The redirect uses window.location.href which bypasses React Router and causes a full page reload
3. The Finance tab promises 'Home Expense' management but delivers nothing

Fix required:
a) Add a 'source' field to Expenses.jsx form with options: home_expense | own_farm | business
b) Embed home expense list directly in Finance.jsx 'home' tab (filter by source='home_expense')

Expenses.jsx

❌ MISSING **Source field is still absent — dashboard expense categorization is broken**

Expenses.jsx has not been updated to include a source field. Every expense is saved without source tagging. The Dashboard tries to filter: expenses where source === 'own_farm' OR !source — the !source fallback means ALL current expenses (which have no source) are counted as business expenses, including what should be home expenses. This inflates the business expense total and understates profit.

```
// Dashboard expense filter — current behaviour:
const gExp = filteredExpenses
  .filter(e => e.source === 'own_farm' || !e.source)  // !e.source catches ALL
current records
  .reduce((sum, e) => sum + (e.amount || 0), 0);

// After adding source field to Expenses.jsx, correct filter should be:
const gExp = filteredExpenses
  .filter(e => e.source !== 'home_expense')  // exclude home from business
  .reduce((sum, e) => sum + (e.amount || 0), 0);
```

Reports.jsx

✔️ WORKS **All 5 report types render correctly with minor data accuracy issues**

Report Type	Status	Known Issue
Monthly Summary	✔️ Works	Harvester income uses item.date which may be missing in older records
Machine-wise	✔️ Works	Only groups by tyre/track; tractor jobs not shown in machine report
Farmer Balance	✔️ Works	Correct outstanding balance per farmer

Driver Payroll	✔️ Works	Shows earned/paid/balance per driver
Seasonal	✔️ Works	Groups by season code, shows job count and income

Settings.jsx

⚠️ ISSUE	**Backup works but has no schemaVersion; restore has no version validation**

Export correctly downloads a JSON backup of all 11 collections. Import uses FileReader to restore. However neither the export nor the import includes a schemaVersion field. The import uses window.confirm() (English) instead of an inline Tamil confirmation. If the schema changes between backups, there is no detection mechanism.

4. Complete Issues List — v2 Audit

4.1 Crash-Level Issues

#	Issue	File	Impact	Fix
C1	generateRentalReceipt not imported in server/index.js	server/index.js	Entire PDF server crashes when Rental PDF button is tapped — server process dies	Add to destructured require: generateRentalReceipt
C2	Rental PDF uses non-existent fields (equipmentName, duration, durationUnit)	server/pdfGenerator.js	Even after fixing C1, the rental PDF renders blank table rows	Replace with rental.machineType, rental.quantity, rental.unit
C3	API URL hardcoded to localhost:5000	src/services/pdfService.js	All PDF generation fails completely when deployed anywhere except local machine	Use environment variable: import.meta.env.VITE_PDF_API_URL

4.2 Logic Errors

#	Issue	File	Impact	Fix
L1	Harvester jobs not filtered by month in dashboard	Dashboard.jsx	Dashboard income and job expenses always show all-time totals regardless	Apply filterByDate(j.date) to filteredJobs

			of month selected	
L2	Default machineType 'tyre' is invalid enum value	DriverEntry.jsx	Logs saved with wrong machine type; breaks Harvester log-linking filter	Change initial machineType to 'harvester_tyre'
L3	Default sessions pre-filled 09:00-10:00	DriverEntry.jsx	Drivers may forget to update; 1 hour logged incorrectly on every entry	Default to empty string: start:'', end:''
L4	Source field missing from Expenses.jsx	Expenses.jsx	Dashboard cannot segregate home vs business expenses; all expenses counted as business	Add source field dropdown (home_expense / own_farm / business)
L5	Finance.jsx Home tab is a broken stub	Finance.jsx	Home expense management feature is non-functional	Embed home expense list using expenses filtered by source='home_expense'
L6	Workers use workerName string not workerId reference	Workers.jsx	Historical entries break if worker name changes; no worker profile link	Save workerId; display worker.name by lookup

4.3 Minor Issues

#	Issue	File	Impact	Fix
M 1	Admin PIN hardcoded as '1234' in source code	Login.jsx	Cannot be changed without modifying code	Store admin password hash in localStorage
M 2	Driver fallback PIN '0000' allows anyone to log in	Login.jsx	Security weakness — any driver can use 0000	Remove the fallback or make it admin-configurable
M 3	Settings backup missing schemaVersion field	Settings.jsx	No version validation on restore; future schema changes undetectable	Add schemaVersion:'2.0.0' to export; validate on import
M 4	alert() / window.confirm() used in SalaryDriver and Settings	SalaryDriver.jsx Settings.jsx	English browser dialogs break Tamil UX consistency	Replace with inline state messages and Tamil text
M	Dead npm	package.json	Increases	Remove from package.json; run

5	dependencies: jspdf, html2pdf.js, pdfmake		frontend bundle size; unused code	npm install
M 6	No start script in server/package.json	server/package.json	Developers must know to run 'node index.js' manually	Add: "start": "node index.js" to scripts
M 7	App.jsx seeding uses getData() for object config	App.jsx	Semantically wrong; works by accident; fragile	Use getConfig('rl_pricing_config') instead
M 8	DriverDashboard has no month filter	DriverDashboard.jsx	Shows all-time salary stats with no period breakdown	Acceptable for now; add month filter in future

5. Changes Needed — High Priority

The three crash-level issues must be fixed before ANY demo or client delivery.
A PDF button crashing the server mid-demo destroys client confidence.
Fix C1, C2, and C3 first — they take under 15 minutes total.

5.1 Fix C1 — Server Import (5 minutes)

```
// server/index.js — Line 3
// CHANGE FROM:
const { generateHarvesterBill, generateStatement } = require('./pdfGenerator');

// CHANGE TO:
const { generateHarvesterBill, generateStatement, generateRentalReceipt }
  = require('./pdfGenerator');
```

5.2 Fix C2 — Rental PDF Field Names (10 minutes)

```
// server/pdfGenerator.js — generateRentalReceipt function
// Find and replace these 3 field references:

rental.equipmentName  →  (rental.machineType || 'Equipment Rental')
rental.duration       →  rental.quantity
rental.durationUnit   →  rental.unit

// Also add farmer village and phone to the header:
doc.text(`Village: ${farmer?.village || 'N/A'}`, 50, topY + 50);

// And add payment history to rental receipt (like harvester bill has):
if (rental.payments && rental.payments.length > 0) {
  doc.moveDown(2);
```

```
  doc.fillColor('#1B3A6B').fontSize(12).text('Payment History');
  rental.payments.forEach(p => {
    doc.fillColor('black').fontSize(10)
      .text(`${p.date} - ${p.mode}: Rs. ${p.amount}`);
  });
}
```

5.3 Fix L1 — Dashboard Harvester Month Filter (2 minutes)

```
// Dashboard.jsx — Replace line:
const filteredJobs = jobs.filter(j => j.status !== 'cancelled');

// With:
const filteredJobs = jobs.filter(j =>
  j.status !== 'cancelled' && filterByDate(j.date)
);

// This also fixes job-level expenses being unfiltered (jExp).
```

5.4 Fix L2 & L3 — DriverEntry Defaults (3 minutes)

```
// DriverEntry.jsx — Fix initial formData state:
const [formData, setFormData] = useState({
  driverId: userId || '',
  machineType: 'harvester_tyre',   // was 'tyre' (invalid enum)
  farmerId: '',
  date: new Date().toISOString().split('T')[0],
  sessions: [{ start: '', end: '', durationHours: 0 }],  // was pre-filled 09:00-
10:00
  diesel: { mode: 'none', value: 0, pricePerLitre: 0, total: 0 }
});

// Also fix the reset after save:
sessions: [{ start: '', end: '', durationHours: 0 }],  // not pre-filled
```

5.5 Fix L4 — Add source Field to Expenses.jsx

```
// Expenses.jsx — Add source field to formData:
const [formData, setFormData] = useState({
  date: new Date().toISOString().split('T')[0],
  source: 'business',   // NEW
  category: 'maintenance',
  amount: '',
  description: ''
});

// Add SelectField before the category dropdown:
<SelectField
  english='Expense Type' tamil='செலவு வகை'
```

```
  options={[
    { value: 'business', ta: 'வியாபார செலவு (Business)' },
    { value: 'own_farm', ta: 'சொந்த பண்ணை செலவு (Own Farm)' },
    { value: 'home_expense', ta: 'வீட்டு செலவு (Home)' },
  ]}
  value={formData.source}
  onChange={(e) => setFormData({...formData, source: e.target.value})}
/>

// Fix Dashboard filter after this change:
const gExp = filteredExpenses
  .filter(e => e.source !== 'home_expense')  // exclude home expenses from profit
  .reduce((sum, e) => sum + (e.amount || 0), 0);
```

5.6 Fix L5 — Finance.jsx Home Expenses Tab

```
// Finance.jsx — Replace the 'home' tab content with actual data:

// In useEffect, add:
const homeExpenses = getData('rl_expenses').filter(e => e.source ===
'home_expense');
setExpenses(homeExpenses);

// Replace the redirect stub with:
{activeTab === 'home' && (
  <div className='list-container'>
    {expenses.map(exp => (
      <div key={exp.id} className='card' style={{borderLeft:'4px solid #C53030'}}>
        <div style={{display:'flex',justifyContent:'space-between'}}>
          <div>
            <div style={{fontWeight:'bold'}}>{exp.description ||
exp.category}</div>
            <div style={{fontSize:'0.8rem',color:'#718096'}}>{exp.date}</div>
          </div>
          <span style={{fontWeight:'bold',color:'#C53030'}}>
            {formatCurrency(exp.amount)}
          </span>
        </div>
      </div>
    ))}
  </div>
)}
```

5.7 Fix M1 & M2 — Admin PIN Security

```
// Login.jsx — Store admin PIN in localStorage instead of hardcoding:

if (role === 'admin') {
  const savedPin = localStorage.getItem('admin_pin') || '1234';
  if (pin === savedPin) {
    onLogin({ role: 'admin', name: 'Administrator' });
  } else {
```

```
    setError('தவறான கடவுச்சொல் (Invalid Admin PIN)');
  }
}

// Remove the '0000' fallback for drivers:
// CHANGE FROM:
if (driver && (driver.pin === pin || pin === '0000')) {

// CHANGE TO:
if (driver && driver.pin === pin) {

// Add a PIN change option in Settings.jsx:
const handleChangeAdminPin = () => {
  const newPin = prompt('Enter new Admin PIN (4 digits):');
  if (newPin && newPin.length === 4 && !isNaN(newPin)) {
    localStorage.setItem('admin_pin', newPin);
    setStatus('PIN updated successfully!');
  }
};
```

5.8 Fix M3 — Backup schemaVersion

```
// Settings.jsx — Add schemaVersion to export:
const backupData = {
  schemaVersion: '2.0.0',
  exportDate: new Date().toISOString().split('T')[0],
  ...   // all collections
};

// Add version check to import:
const data = JSON.parse(event.target.result);
if (!data.schemaVersion) {
  setStatus('Invalid backup: missing version. (பதிப்பு விவரம் இல்லை)');
  return;
}
if (data.schemaVersion !== '2.0.0') {
  setStatus(`Incompatible backup version: ${data.schemaVersion}`);
  return;
}
// Then restore...
```

5.9 Fix M5 — Remove Dead Dependencies

```
# Remove unused frontend PDF libraries:
npm uninstall jspdf html2pdf.js pdfmake

# Verify package.json dependencies section should only contain:
#  react, react-dom, react-router-dom, vite-plugin-pwa
```

5.10 Fix M6 — Server Start Script

```
// server/package.json — add start script:
{
  "scripts": {
    "start": "node index.js",
    "dev": "node --watch index.js"
  }
}

// Now server can be started with: npm start (from server/ directory)
```

6. Tomorrow's Action Plan

Time available: 1 day (before April 25 deadline).
Priority: Stability first, then correctness, then polish.
Do NOT add new features — fix what exists.
Test on mobile after every fix.

Phase 1 — Crash Fixes (30 minutes)

These must be done first. A crashing PDF server means the client cannot see bills.

Step	Task	File	Time
1	Fix generateRentalReceipt import in server/index.js	server/index.js	5 min
2	Fix rental PDF field names (machineType, quantity, unit)	server/pdfGenerator.js	10 min
3	Test all 3 PDF endpoints manually — Harvester, Rental, Statement	Browser + server	10 min
4	Add environment variable support for API_BASE_URL	pdfService.js	5 min

Phase 2 — Logic Fixes (1.5 hours)

Core calculation accuracy. The client will notice if numbers are wrong.

Step	Task	File	Time
5	Fix Dashboard: apply month filter to harvesterJobs	Dashboard.jsx	10 min

6	Add source field to Expenses.jsx form	Expenses.jsx	15 min
7	Fix Finance.jsx Home tab: show home expenses inline	Finance.jsx	15 min
8	Fix DriverEntry default machineType and sessions	DriverEntry.jsx	5 min
9	Fix Workers.jsx: workerName → workerId with lookup	Workers.jsx	20 min
10	Test Dashboard with real data: verify income/expense totals	Browser	15 min

Phase 3 — Polish & Security (1 hour)

Client-facing quality. This is what makes the system feel professional.

Step	Task	File	Time
11	Store admin PIN in localStorage (not hardcoded)	Login.jsx + Settings.jsx	10 min
12	Remove driver fallback PIN '0000'	Login.jsx	3 min
13	Replace alert() in SalaryDriver with Tamil inline message	SalaryDriver.jsx	5 min
14	Add schemaVersion to backup export + validation on import	Settings.jsx	15 min
15	Add npm start script to server/package.json	server/package.json	2 min
16	Remove jspdf, html2pdf.js, pdfmake from package.json	package.json	5 min
17	Fix App.jsx seeding: getData → getConfig for pricing config	App.jsx	2 min
18	Full mobile test: login as driver, log entry, salary view	Mobile browser	20 min

Phase 4 — Final Verification (30 minutes)

Test Scenario	Expected Result	Who Tests
Farmer logs in → taps Rental PDF	PDF downloads with correct machine and quantity	Admin verifying farmer view
Admin logs in → Dashboard April 2026	Only April jobs and rentals shown in income	Admin with test data from multiple months
Driver logs entry with harvester_tyre	Log saved with correct machineType; visible in	Driver on mobile

	Harvester log-linker	
Add home expense → check Finance tab	Shows in Finance → Home Expenses list	Admin
Export backup → delete a record → restore	All records restored; schemaVersion validated	Admin
Admin PIN change → logout → re-login with new PIN	New PIN works; old PIN rejected	Admin

7. Improvement Suggestions

7.1 UX Improvements for Rural Low-Literacy Users

- Replace all window.alert() and window.confirm() calls with Tamil inline messages and in-app confirmation dialogs. Browser dialogs are in English and unfamiliar to rural users.
- Add Tamil success toast notifications after every save action — currently some modules show success, others show nothing.
- Add a prominent 'Logout' indicator showing the current user role in Tamil: 'நிர்வாகி' / 'ஓட்டுநர்' / 'விவசாயி'.
- Increase minimum touch target height to 48px for all buttons and interactive elements — critical for field use on low-cost Android phones.
- Consider adding large icon indicators (harvester icon, tractor icon) next to machine type dropdowns for low-literacy driver users.

7.2 Tamil Consistency Audit

- Harvester.jsx label 'Other Expenses (இதர செலவுகள்)' uses label/amount free text — the label field is not Tamil-guided. Add a category dropdown with Tamil options.
- Reports page table headers (Month, Income, Expense, Profit) are English only — add Tamil headers below each English header.
- Settings page 'App Info' section says 'Version: 2.0.0 (Correction Patch)' — update to 3.0 and add Tamil: 'பதிப்பு: 3.0'.

7.3 Performance Optimization

- Remove jspdf, html2pdf.js, and pdfmake from package.json — they are dead code inflating the frontend bundle.
- The vfs_fonts.js and tamilFont.js files in src/assets/fonts/ contain Base64 font data that is no longer used (font is now on the server). Remove these files to reduce bundle size.
- Consider moving the PDF server to a lightweight cloud service (Railway.app or Render free tier) so the client does not need to run a local Node.js server for PDF generation.

7.4 Future Scalability

- Add a confirmation screen before creating a Harvester Job showing all linked logs and computed totals — prevents accidental duplicate billing.
- Add an 'Edit Harvester Job' capability for Admin (currently jobs can only be viewed, not corrected after save).
- Implement the planned cloud sync feature — replace storage.js LocalStorage calls with API calls when backend is available, with zero frontend changes required.
- Add a Notes field to Driver Logs for the driver to record issues or special conditions on that day.

8. Current System Status Summary

Module	Overall Status	Usable for Client?	Priority to Fix
Authentication (Login)	✔️ Working	Yes — with minor security notes	Low
Admin Dashboard	⚠️ Month filter bug for Harvester	Partially — income figures wrong	HIGH
Driver Entry	⚠️ Wrong defaults	Partially — time/machine type issues	HIGH
Driver Dashboard	✔️ Working	Yes	Low
Driver Salary Ledger	✔️ Working	Yes	Low
Farmer View	⚠️ Sort issue; Rental PDF crash	Partially — rental PDF unavailable	CRITICAL
Harvester Management	✔️ Working	Yes	Low
Machinery Rental	✔️ Working (PDF crashes)	Partially — PDF unavailable	CRITICAL
Worker Management	⚠️ Name vs ID issue	Partially	Medium
Finance (Lending)	✔️ Working	Yes	Low
Finance (Home Expenses)	❌ Non-functional stub	No	HIGH
Own Farm Income	✔️ Working	Yes	Low
Expenses	⚠️ No source field	Partially — dashboard figures wrong	HIGH
Reports	✔️ Working	Yes — with minor data gaps	Low
Settings (Backup)	⚠️ No schemaVersion	Partially	Medium

PDF — Harvester Bill	✔️ Working	Yes	None
PDF — Account Statement	✔️ Working	Yes	None
PDF — Rental Receipt	❌ Server crashes	No	CRITICAL
PWA Install	✔️ Working	Yes	None

END OF DOCUMENTATION — VERSION 3.0

SmartUzhavan — கிராம கணக்கு | 3 Crash Fixes | 6 Logic Fixes | 5 Minor Fixes